SIMATS ENGINEERING

ASSESSMENT TOOL -3

COURSE NAME: COMPUTER NETWORKS

COURSE CODE: CSA0706

COURSE OUTCOME COVERED

CO2: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. (BL3)

Assessment Tool Weightage: 30%

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Annexure A

Debugging & Optimisation Task

Code of Conduct:

I D.Uday Kiran (192425161) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

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1. A rapidly growing IT company has shifted its operations to a smart office building. Employees are facing WLAN and WWAN connectivity issues. As a network engineer, identify the problems, debug the network, and suggest optimization techniques.

Answer

1. Introduction

The company uses a Wireless Local Area Network (WLAN) for internal communication and Wireless Wide Area Network (WWAN) services for remote employees. Due to increased users and device connectivity, the network experiences performance, connectivity, and security issues.

A systematic debugging and optimization approach is required to improve wireless performance, reliability, and security.

2. Problem Identification (Root Cause Analysis)

The major problems identified are:

Issue	Root Cause
Slow internet speed during peak hours	High number of connected devices causing bandwidth congestion
Frequent Wi-Fi disconnection	Poor roaming configuration between floors
Weak signals in conference rooms	Improper access point placement
Video conferencing freezing	High latency, packet loss, and insufficient QoS
AP overload	Unequal distribution of wireless clients
WWAN connection loss in rural areas	Weak cellular coverage
Security threats	Usage of outdated wireless security protocols

3. Debugging Approach

Step 1: Monitor Network Performance

Tools:

- Wi-Fi analyzer
- Network monitoring software
- Signal strength monitoring tools

Purpose:

- Identify overloaded access points
- Analyze bandwidth usage
- Detect interference

Step 2: Analyze Wireless Coverage

Check:

- Signal strength
- Dead zones
- Channel interference

Solution:

- Reposition access points
- Increase coverage in weak areas

Step 3: Check Access Point Utilization

Problem:

Some APs are overloaded while others are unused.

Debugging:

- Analyze connected clients
- Monitor traffic load

Solution:

Enable:

- Load balancing
- Automatic client distribution

Step 4: Perform Security Audit

Problem:

Old security protocols expose the network to attacks.

Solution:

Upgrade:

- WPA2/WPA3 Enterprise security
- Strong authentication methods

- Encryption policies

4. Optimization Techniques

WLAN Optimization

1. Deploy additional access points

Benefits:

- Improves coverage
- Reduces congestion

2. Implement band steering

- Moves users from crowded 2.4 GHz band to 5 GHz band.
- Improves speed and reduces interference.

3. Enable fast roaming protocols:

- IEEE 802.11k
- IEEE 802.11r
- IEEE 802.11v

Advantages:

- Seamless movement between floors
- Reduced connection drops

Traffic Optimization

Implement Quality of Service (QoS).

Priority allocation:

Application	**Priority**
Video conferencing	High
Voice communication	High
File sharing	Medium
Web browsing	Low

WWAN Optimization

Solutions:

- Upgrade to 5G-enabled devices

- Use multiple network providers
- Enable automatic network switching
- Install external antennas in weak coverage areas

5. Analysis and Justification

The major cause of network problems is insufficient wireless capacity planning and poor traffic management.

By implementing:

- AP load balancing
- QoS policies
- Better wireless security
- Improved WWAN coverage

the network can achieve higher speed, reduced latency, and reliable communication.

6. Result / Expected Outcome

After optimization:

- Wi-Fi coverage improves across all floors.
- Network congestion decreases.
- Video conferencing becomes stable.
- Wireless security is strengthened.
- Remote employees receive better WWAN connectivity.

2. A multinational financial organization operates its headquarters, regional offices, and data centers using an ATM network. The organization is facing delays in transactions, voice quality issues, and video conferencing problems. As a network engineer, identify the faults, debug the ATM network, and recommend optimization techniques.

Answer

1. Introduction

The financial organization uses an **Asynchronous Transfer Mode (ATM)** network to support mission-critical applications such as online banking, Voice over IP (VoIP), video conferencing, and real-time financial transactions.

ATM networks use fixed-size **53-byte cells** and Virtual Circuits (VCs) to provide predictable Quality of Service (QoS). Due to increased traffic during peak hours, the network experiences congestion, delay, and inefficient resource utilization.

A proper debugging and optimization process is required to restore reliable and high-performance communication.

2. Problem Identification (Root Cause Analysis)

The major issues identified are:

Problem	Root Cause
Delay in online transactions	ATM switch congestion and high traffic load
Voice call jitter and packet loss	Improper QoS configuration and VC congestion
Poor video conferencing quality	Insufficient bandwidth allocation
High ATM switch utilization	Excessive traffic concentration
Congested Virtual Circuits	Improper VC configuration
Inefficient bandwidth usage	Incorrect ATM service category assignment

3. Debugging Approach

Step 1: Monitor ATM Switch Performance

Network engineers analyze:

- Switch utilization
- Cell loss rate
- Traffic flow
- Delay measurements

Purpose:

- Identify overloaded switches
- Detect congestion points

Step 2: Analyze Virtual Circuit (VC) Configuration

Problem:

Some Virtual Circuits are overloaded.

Debugging process:

- Check active VCs
- Measure bandwidth consumption

- Identify congested paths

Solution:

- Reconfigure VCs
- Distribute traffic across multiple paths

Step 3: Check ATM Service Categories

ATM provides different service categories:

Service Category	**Application**
CBR (Constant Bit Rate)	Voice and real-time communication
VBR (Variable Bit Rate)	Video applications
ABR (Available Bit Rate)	Data applications
UBR (Unspecified Bit Rate)	Non-critical traffic

Problem:

Applications are using unsuitable service categories.

Solution:

Assign proper categories based on application requirements.

Step 4: Analyze Bandwidth Allocation

Problem:

Available bandwidth is not efficiently distributed.

Debugging:

- Monitor traffic patterns
- Identify bandwidth-consuming applications

Solution:

Apply dynamic bandwidth management.

4. Optimization Techniques

1. Implement Quality of Service (QoS)

Traffic prioritization:

Traffic Type	**Priority**

Traffic Type	Priority
Financial transactions	Very High
Voice communication	High
Video conferencing	High
General data transfer	Medium

Benefits:

- Reduced delay
- Improved reliability
- Better user experience

2. Optimize Virtual Circuits

Techniques:

- Create additional Virtual Circuits
- Balance traffic among available paths
- Remove unnecessary VCs

Advantages:

- Reduces congestion
- Improves network efficiency

3. Upgrade ATM Switch Capacity

Solutions:

- Increase switch processing capability
- Add additional switching resources
- Replace overloaded devices

Benefits:

- Handles higher traffic volume
- Reduces cell loss

4. Proper ATM Service Category Assignment

Recommended allocation:

Application	ATM Category
Online banking	CBR/VBR
VoIP	CBR
Video conferencing	VBR
File transfer	ABR

5. Analysis and Justification

The main reason for ATM performance degradation is improper traffic management and inefficient allocation of network resources.

By applying:

- QoS mechanisms
- Correct service category selection
- VC optimization
- Bandwidth management

the ATM network can provide predictable delay, low jitter, and reliable communication.

6. Result / Expected Outcome

After optimization:

- Transaction delays are reduced.
- Voice quality improves.
- Video conferencing becomes stable.
- ATM switches operate efficiently.
- Bandwidth utilization increases.
- Cell loss probability decreases.

3. A multinational e-commerce company operates its headquarters, regional offices, warehouses, and cloud data centers using an MPLS network. The company is experiencing slow cloud access, delayed order processing, poor voice quality, and video conferencing interruptions. As a network engineer, identify the faults, debug the MPLS network, and recommend optimization techniques.

Answer

1. Introduction

The e-commerce company uses a **Multiprotocol Label Switching (MPLS)** network to connect multiple locations securely and efficiently. MPLS provides fast packet forwarding using labels instead of traditional IP-based routing.

The network supports critical applications such as:

- Online order processing
- Enterprise Resource Planning (ERP)
- Cloud inventory management
- Voice over IP (VoIP)
- Video conferencing
- Customer support systems

Due to increased network traffic, the company experiences congestion, latency, and inefficient routing problems. A systematic debugging and optimization approach is required to improve MPLS performance.

2. Problem Identification (Root Cause Analysis)

The major issues identified are:

Problem	**Root Cause**
Slow access to cloud applications	Congestion in MPLS paths and insufficient bandwidth
Delayed order processing	High network latency and overloaded routes
Poor voice call quality	Packet delay, jitter, and improper QoS configuration
Video conferencing interruptions	Bandwidth congestion and packet loss
Some LSPs are overloaded	Unequal traffic distribution
Incorrect label mappings	Faulty MPLS label configuration
Inefficient bandwidth utilization	Lack of traffic engineering

3. Debugging Approach

Step 1: Analyze MPLS Label Configuration

MPLS forwards packets using labels assigned by routers.

Debugging process:

- Verify Label Forwarding Information Base (LFIB)
- Check label mappings
- Identify incorrect label assignments

Tools used:

- MPLS traceroute
- Router diagnostic commands
- Network monitoring systems

Solution:

- Correct incorrect label mappings
- Reconfigure Label Distribution Protocol (LDP)

Step 2: Monitor Label Switched Paths (LSPs)

Problem:

Some LSPs are congested while others are underutilized.

Debugging:

Analyze:

- LSP bandwidth usage
- Traffic flow
- Delay measurements

Solution:

- Redistribute traffic
- Create optimized paths

Step 3: Check Traffic Engineering Configuration

Problem:

Traffic engineering is not properly configured.

Debugging:

Check:

- Available bandwidth

- Route selection
- Path utilization

Solution:

Enable:

- MPLS Traffic Engineering (MPLS-TE)

Benefits:

- Efficient bandwidth usage
- Reduced congestion
- Better load balancing

Step 4: Analyze QoS Configuration

Problem:

Critical applications are not receiving proper priority.

Check:

- Voice traffic priority
- Video traffic priority
- Data traffic allocation

Solution:

Implement QoS policies.

4. Optimization Techniques

1. MPLS Traffic Engineering Optimization

MPLS-TE allows network administrators to control traffic paths.

Advantages:

- Prevents overloaded links
- Uses unused bandwidth
- Improves network efficiency

2. Load Balancing Across LSPs

Current problem:

Some paths are overloaded.

Optimization:

- Distribute traffic among multiple LSPs
- Select alternate paths during congestion

Result:

- Reduced delay
- Improved throughput

3. Quality of Service (QoS) Implementation

Traffic prioritization:

Application	**Priority**
VoIP	Very High
Video Conferencing	High
Order Processing	High
Cloud Applications	Medium
General Data	Low

Benefits:

- Reduced latency
- Improved communication quality
- Better application performance

4. Correct MPLS Label Management

Optimization steps:

- Update label tables
- Verify LDP operation
- Remove incorrect mappings

Benefits:

- Faster packet forwarding
- Reduced routing errors

5. Bandwidth Optimization

Techniques:

- Upgrade overloaded links
- Monitor bandwidth usage
- Apply efficient traffic policies

Benefits:

- Better resource utilization
- Improved network scalability

5. Analysis and Justification

The main reason for MPLS network performance degradation is improper traffic distribution, incorrect label mappings, and lack of traffic engineering.

By applying:

- MPLS-TE
- QoS policies
- LSP optimization
- Label correction

the network can achieve faster packet forwarding, reduced congestion, and improved reliability.

6. Result / Expected Outcome

After optimization:

- Cloud applications respond faster.
- Order processing delays decrease.
- Voice and video quality improve.
- Network bandwidth is utilized efficiently.
- MPLS paths become balanced.
- Overall network reliability increases.

4. A multinational logistics company operates an automated warehouse where barcode scanners, RFID readers, autonomous robots, and inventory systems exchange data through a computer network. The company is facing data transmission errors, incorrect records, and failed barcode scans. As a network engineer, identify the errors, debug the communication system, and recommend optimization techniques.

Answer

1. Introduction

The logistics company uses an automated warehouse network where thousands of data transactions occur every day between:

- Barcode scanners
- RFID readers
- Autonomous robots
- Inventory management systems
- Central database servers

Reliable data communication is essential because incorrect or lost information can cause inventory mismatches, shipment delays, and operational failures.

The network engineer must analyze transmission errors, improve error detection and correction mechanisms, and optimize the communication system.

2. Problem Identification (Root Cause Analysis)

The major issues identified are:

Problem	**Root Cause**
Incorrect shipment records	Undetected data corruption during transmission
Barcode scan failures	Signal interference and communication errors
Inventory mismatch	Loss of data integrity
Increased retransmissions	High frame error rate
Corrupted frames	Electrical interference and signal attenuation
Network congestion	Excessive traffic during peak hours
Some errors remain undetected	Weak error detection mechanisms

3. Debugging Approach

Step 1: Analyze Transmission Errors

The network engineer monitors:

- Frame error rate

- Packet loss
- Retransmission count
- Signal quality

Tools used:

- Network analyzers
- Packet monitoring tools
- Error logs

Purpose:

- Identify error locations
- Detect communication failures

Step 2: Check Error Detection Mechanisms

Existing problem:

Some corrupted frames are not detected.

The current error detection methods are analyzed:

Parity Check

Limitation:

- Detects only simple bit errors.
- Cannot detect multiple-bit errors effectively.

Checksum

Limitation:

- Less reliable for high-error environments.

Cyclic Redundancy Check (CRC)

Solution:

Implement CRC because it provides:

- High error detection capability
- Better data integrity
- Reliable frame checking

Step 3: Analyze Physical Layer Issues

Problem:

Transmission errors occur due to:

- Electrical interference
- Signal attenuation
- Weak wireless signals

Debugging:

Check:

- Cable quality
- Signal strength
- Network hardware condition

Solutions:

- Replace damaged cables
- Reduce electromagnetic interference
- Improve wireless coverage

Step 4: Monitor Network Congestion

Problem:

During peak hours, increased traffic causes transmission delays.

Debugging:

Analyze:

- Network utilization
- Traffic patterns
- Device communication rate

Solution:

Implement traffic management techniques.

4. Optimization Techniques

1. Implement Advanced Error Detection

Use:

Cyclic Redundancy Check (CRC)

Advantages:

- Detects burst errors
- Improves data accuracy
- Reduces incorrect database entries

2. Apply Error Correction Techniques

Implement:

Automatic Repeat Request (ARQ)

Working:

1. Sender transmits data.
2. Receiver checks errors.
3. If error occurs, receiver requests retransmission.
4. Correct data is received.

Benefits:

- Reliable communication
- Prevents corrupted data storage

3. Improve Signal Quality

Optimization methods:

- Use shielded cables
- Increase signal strength
- Replace faulty network equipment
- Reduce interference sources

Benefits:

- Fewer transmission errors
- Stable communication

4. Optimize Network Traffic

Solutions:

- Separate IoT device traffic using VLANs
- Apply Quality of Service (QoS)

- Increase network bandwidth

Benefits:

- Reduced congestion
- Faster data transmission

5. Regular Network Monitoring

Implement:

- Real-time error monitoring
- Automated alerts
- Performance analysis tools

Benefits:

- Early fault detection
- Reduced downtime

5. Analysis and Justification

The primary cause of warehouse communication failures is unreliable error control combined with physical interference and network congestion.

By implementing:

- CRC-based error detection
- ARQ-based error correction
- Signal improvement methods
- Traffic optimization

the network can maintain accurate and reliable data exchange.

These techniques ensure that inventory records, shipment details, and barcode information remain correct.

6. Result / Expected Outcome

After optimization:

- Data transmission accuracy improves.
- Barcode scanning becomes reliable.
- Inventory mismatches reduce.
- Retransmission overhead decreases.

- Network performance improves.
- Warehouse automation becomes more efficient.

5. A leading online education platform conducts live classes and online examinations for more than 15,000 students. The platform faces incomplete downloads, delayed responses, duplicate packets, retransmissions, and receiver buffer overflow problems. As a network engineer, identify the issues, debug the communication system, and recommend optimization techniques.

Answer

1. Introduction

The online education platform provides live classes, online examinations, and digital learning resources to thousands of students simultaneously.

The system involves:

- Central high-speed server
- Student devices
- Broadband and mobile networks
- Data transmission protocols

During peak usage, the server sends large amounts of data such as:

- Question papers
- Images
- Videos
- Answer acknowledgements

Due to differences in network speeds between the server and students, communication problems occur. Proper flow control and optimization techniques are required to ensure reliable data transfer.

2. Problem Identification (Root Cause Analysis)

The major problems identified are:

Problem	**Root Cause**
Incomplete downloads	Sender transmits data faster than receiver capacity
Delayed page loading	High network latency and congestion
Duplicate packets	Excessive retransmissions

Problem	Root Cause
Frequent retransmissions	Packet loss and improper flow control
Receiver buffer overflow	Sender does not regulate transmission rate
Reduced throughput	Inefficient bandwidth utilization
Increased latency	Network congestion during peak hours

3. Debugging Approach

Step 1: Analyze Network Traffic

The network administrator monitors:

- Packet transmission rate
- Packet loss
- Round Trip Time (RTT)
- Throughput
- Buffer utilization

Tools used:

- Network monitoring software
- Packet analyzers
- Performance monitoring systems

Purpose:

- Identify congestion points
- Measure communication efficiency

Step 2: Check Flow Control Mechanism

Existing Issue:

The sender transmits data faster than the receiver can process.

This causes:

- Buffer overflow
- Packet loss
- Retransmissions

Debugging:

Analyze:

- Receiver window size
- Data transmission rate
- Acknowledgement delays

Step 3: Analyze TCP Window Management

The platform uses TCP communication.

Problem:

Fixed or improper window size causes inefficient data transfer.

Solution:

Enable:

TCP Sliding Window Protocol

Working:

1. Sender transmits multiple packets without waiting for every acknowledgement.
2. Receiver advertises available buffer space.
3. Sender adjusts transmission rate according to receiver capacity.

Benefits:

- Prevents buffer overflow
- Improves throughput

Step 4: Monitor Congestion Control

Problem:

Large number of students accessing the server simultaneously causes congestion.

Debugging:

Check:

- Server bandwidth utilization
- Network traffic load
- Packet retransmission rate

Solution:

Apply congestion control algorithms.

4. Optimization Techniques

1. Implement Sliding Window Flow Control

Advantages:

- Controls sender transmission speed
- Matches sender and receiver capacity
- Reduces packet loss

2. Dynamic Window Size Adjustment

Technique:

Use TCP congestion window adjustment.

Benefits:

- Increases speed when network is stable
- Reduces traffic during congestion
- Improves reliability

3. Server Load Optimization

Solutions:

Deploy Content Delivery Network (CDN)

Benefits:

- Reduces server load
- Provides faster content delivery
- Reduces latency for students

Use Multiple Servers

Advantages:

- Handles large user requests
- Prevents single server overload

4. Bandwidth Management

Implement:

Quality of Service (QoS)

Priority allocation:

Traffic Type	**Priority**
Examination data	Very High
Live classes	High
Video content	Medium
General downloads	Low

Benefits:

- Reduces delays
- Ensures important traffic delivery

5. Packet Retransmission Optimization

Problems:

Excessive retransmissions waste bandwidth.

Solutions:

- Improve error detection
- Optimize TCP timeout values
- Reduce packet loss

Benefits:

- Higher throughput
- Lower latency

5. Analysis and Justification

The main reason for communication failure is the mismatch between the high-speed server and slower student networks.

The server sends data faster than receivers can process, resulting in:

- Buffer overflow
- Packet loss
- Retransmissions

By implementing:

- Sliding window protocol

- TCP congestion control
- CDN deployment
- QoS policies

the system can dynamically adjust transmission speed and provide reliable communication.

6. Result / Expected Outcome

After optimization:

- Download failures reduce.
- Page loading becomes faster.
- Packet retransmissions decrease.
- Buffer overflow is prevented.
- Network throughput improves.
- Online examinations run smoothly.